

Supplementary Material: Numerical-consistency and literature and physical benchmark validations

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This supplement documents the validation suite that underpins the valley-aware spin-shuttling simulator used in the main text. The goal is not to establish new physics but to certify that (i) the numerical engine reproduces analytic limits to machine precision, and (ii) the representative parameter values quoted in the main text are consistent with independent literature results. We group the checks into *numerical-consistency* validations (Sec. S1) and *literature and physical benchmark* validations (Sec. S2). Every check is reproducible from the released code with a fixed seed; the relevant script is named in each case. Throughout, “relative error” means $|x_{\text{num}} - x_{\text{ref}}|/|x_{\text{ref}}|$.

Two cautions apply globally. First, several checks are validated only in a *ratio* or *trend* sense; the accompanying caveats state exactly what is and is not claimed. Second, these validations concern the spin and field sectors and the generic valley two-level dynamics; they do not validate the phenomenological spin–valley coupling ansatz, which by construction is a diagnostic choice rather than a microscopically derived form (see the main-text Model section).

S1. NUMERICAL-CONSISTENCY VALIDATIONS

These checks compare the simulator against closed-form expressions or against automatic differentiation, in regimes where an exact answer is known.

A. Analytic field landscape (V0)

The deterministic field model reproduces the analytic periodic landscape of the device model to machine precision: the analytic derivative $\partial_x B_{\text{long}}$ and the JAX automatic-differentiation value agree to a relative error of 4×10^{-9} (`field_landscape.py`). This certifies the field gradients that feed every downstream observable.

B. Bounded deterministic modulation phase (V0.1)

For uniform-velocity shuttling through a periodic landscape, the maximum deterministic modulation phase $|\phi_{\text{mod}}|_{\text{max}}$ is independent of the number of shuttling periods N , whereas for a monotonic-gradient landscape it grows as $\propto N^2$. Over $N = 1$ –100 the periodic-landscape relative variance is 0.00%, while the monotonic case grows from ~ 4 to $\sim 4.4 \times 10^4$ radians. *Caveat:* this is a property of the *deterministic* phase, which is a calibration target, not an error; the noise-driven variance is treated in the literature-reproduction checks below.

C. Quasi-static charge-noise limit (V0.2)

For a static dot with Gaussian quasi-static position fluctuations and a monotonic gradient, the simulator reproduces the analytic dephasing time $T_2^* = \sqrt{2}/\sigma_\omega$ to within 1%: a 2×10^4 -realization Monte Carlo gives 27.1 ns against the predicted 26.8 ns.

D. Double precision is required (V0.3)

The mixed nanometre and tesla-per-metre scales make single precision insufficient for the derivative and cancellation steps used here. In float32 the analytic and automatic-differentiation derivatives disagree at the $\sim 10^0$ T/m level; in float64 they agree at $\sim 10^{-9}$ T/m (`reproduce/v0p3_float_precision_check.py`). All production runs therefore use 64-bit precision (`jax_enable_x64`). This also guarantees the accuracy of the geometry-derivative checks below.

TABLE S1. Summary of numerical-consistency validations. “Tolerance” is the relative error against the analytic or finite-difference reference unless noted.

ID	quantity	reference	result
V0	field-gradient autodiff	analytic derivative	4×10^{-9}
V0.1	modulation-phase scaling	N -independence / N^2	0.00% variance
V0.2	quasi-static T_2^*	$\sqrt{2}/\sigma_\omega$	$< 1\%$ (27.1 vs 26.8 ns)
V0.3	float64 necessity	analytic vs autodiff	$10^0 \rightarrow 10^{-9}$
V6	geometry autodiff	finite difference	$\leq 4 \times 10^{-6}$
V7	array periodicity	$B(x+a)=B(x)$	0.04%
V8	harmonic content	FFT spectrum	$N_{\text{harm}}=3$ retained

E. Position and geometry autodifferentiation (V6)

Automatic differentiation of B_z, B_x with respect to both position *and* device-geometry parameters (magnet half-extents $\text{half}_x, \text{half}_z$, spatial period, 2DEG depth) agrees with finite differences to a relative error $\leq 4 \times 10^{-6}$; position derivatives reach $\sim 10^{-10}$, and the four geometry derivatives are 1.3×10^{-6} (half_x), 2.0×10^{-6} (half_z), 1.0×10^{-6} (period), and 4.0×10^{-6} (depth) (`tests/test_geometry_autodiff.py`). This underpins any differentiable geometry exploration.

F. Spatial periodicity of the finite array (V7)

An eleven-prism finite array (central ± 5 periods) satisfies $B(x+a) = B(x)$ to within 0.04% in the central region (maximum deviation 0.0423% at $z = -50$ nm, $a = 400$ nm), justifying the use of a periodic descriptor.

G. Fourier-harmonic content (V8)

The stray field is dominated by its first harmonic, but the even/odd harmonic pattern depends on the magnet duty cycle. For a large magnet (200³ nm, $a = 400$ nm, 50% duty) only odd harmonics survive, with $B_z^{(3)}/B_z^{(1)} = 8.8\%$ and $B_x^{(3)}/B_x^{(1)} = 10.0\%$; for the small-magnet baseline (50³ nm, $a = 150$ nm, 33% duty) even harmonics also appear, with $B_z^{(2)}/B_z^{(1)} = 9.2\%$. We therefore retain three harmonics ($N_{\text{harm}} = 3$) in the field descriptor so that both regimes are represented. Table S1 summarizes the numerical-consistency checks of this section.

S2. LITERATURE AND PHYSICAL BENCHMARK VALIDATIONS

These checks reproduce independently published results in the regime where they apply, and motivate the representative parameter values used in the main text. Each is validated in the sense stated; absolute magnitudes that depend on arbitrary noise amplitudes are validated only as ratios.

A. Gradient-halving dephasing ratio (V1)

Halving a monotonic magnetic gradient roughly doubles the static-dot T_2^* , consistent with the ratio reported by Krzywda *et al.* [1], who observe $T_2^* = 4.4 \rightarrow 8.5 \mu\text{s}$ on reducing the longitudinal gradient, a ratio of 1.93. At $g_{\text{grad}} = 0.1 \rightarrow 0.05 \text{ mT/nm}$ we find $T_2^* = 302 \rightarrow 664 \text{ ns}$, a ratio of 2.20. Our acceptance interval is $\pm 20\%$ of the reported ratio, that is [1.55, 2.32]; the interval is our own tolerance, not a range reported in the reference. *Caveat:* the absolute values are about $10\times$ shorter than the reference because our noise amplitude $\sigma_{\delta x} = 0.3 \text{ nm}$ is a representative, not device-calibrated, value; only the ratio is claimed. Figure S1 shows the two decay curves.

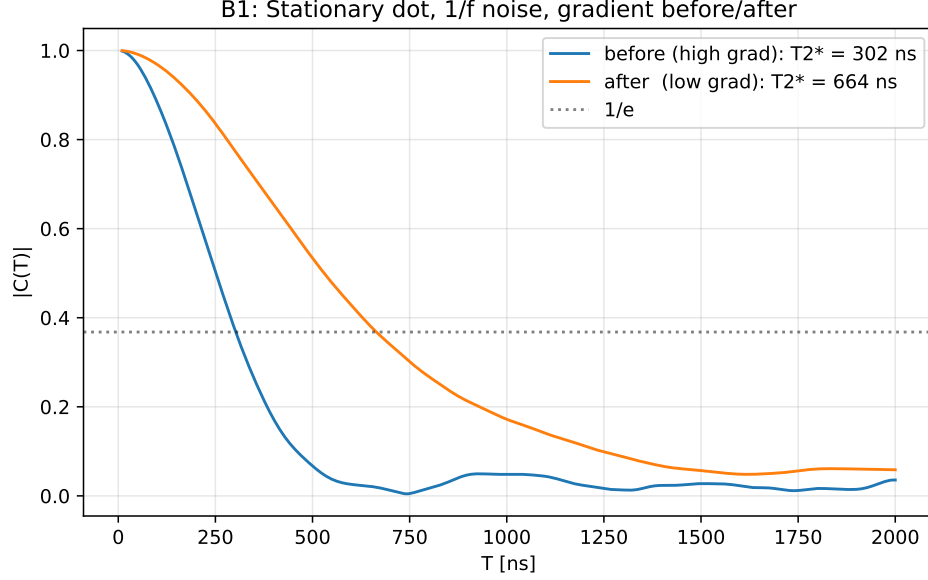


FIG. S1. Validation V1. Static-dot coherence decay $|C(T)|$ under $1/f$ charge noise for a high and a halved magnetic gradient. Halving the gradient increases T_2^* from 302 to 664 ns (ratio 2.20), within our $\pm 20\%$ acceptance interval around the reported ratio. Only the ratio is claimed; the absolute values depend on the representative noise amplitude.

B. Motional-narrowing trend (V2)

For a periodic landscape with $1/f$ charge noise, the effective T_2^* improves as the shuttling velocity increases, from 50 ns at $v = 0$ to 3215 ns at $v = 20$ m/s (up to $64\times$), reproducing the direction of the motional-narrowing improvement reported in Ref. [1]; the $64\times$ magnitude is specific to the present model and parameter set. This occurs once the drive frequency $f_{\text{drive}} = v/a$ moves above the dominant noise band. The stationary baseline here is an effective static-gradient model with the mean absolute periodic gradient $\langle |B'_{\text{long}}| \rangle \approx 0.8 \text{ T}/\mu\text{m}$, so only the *relative* improvement is meaningful. *Caveat:* we claim only the direction of the trend; at very high velocity, orbital non-adiabaticity (not modeled here) is expected to re-introduce error, so unbounded improvement is not claimed.

C. Filter-function suppression (V3)

The periodic-shuttling filter function $F(f, T)$ suppresses low-frequency ($f < 1 \text{ MHz}$) sensitivity by about 2×10^5 relative to a static dot. More directly, the actual dephasing functional, the PSD-weighted overlap $\chi(T) = \int S_{\delta x}(f) F(f, T) df$, is suppressed by about 5.8×10^3 ($T = 1 \mu\text{s}$, $v = 10 \text{ m/s}$). *Caveat:* the filter-function shape and the realized dephasing differ; a 10^5 -fold suppression of F corresponds to only a 10^3 -fold suppression of χ , because dynamical decoupling lowers the low-frequency response but introduces a sensitivity peak near f_{drive} that overlaps the $1/f$ tail. Figure S2 shows the filter function and the PSD-weighted integrand.

D. Valley Landau-Zener dynamics (V4)

The two-level valley time evolution reproduces the analytic Landau-Zener diabatic probability

$$P_{\text{diab}} = \exp\left(-\frac{2\pi\Delta_v^2}{\hbar|\dot{\epsilon}_v|}\right) \quad (\text{S1})$$

(where Δ_v is the valley avoided-crossing coupling and ϵ_v the diabatic valley detuning; Fig. S3) to within 0.28% over a sweep-rate parameter range $\gamma \in [0.1, 10]$. In the adiabatic limit ($v \rightarrow 0$) deep-pocket traversal gives an excitation probability $P_e < 10^{-5}$, and in the diabatic limit ($v \geq 30 \text{ m/s}$) the numerical and analytic predictions agree to order of magnitude. *Caveat:* this validates the *reduced* valley Landau-Zener physics of Oda *et al.* [2], not their full optimized-protocol sequence; at intermediate velocity ($1 \leq v \leq 10 \text{ m/s}$) Stückelberg interference produces

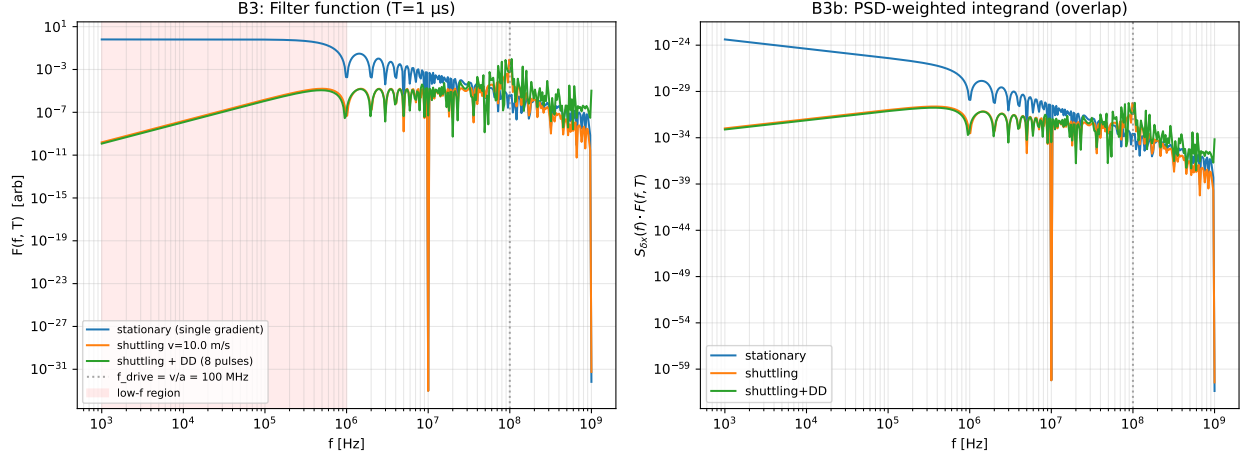


FIG. S2. Validation V3. Left: the filter function $F(f, T)$ for a static dot versus periodic shuttling (and shuttling with dynamical decoupling), showing strong low-frequency ($f < 1$ MHz, shaded) suppression. Right: the PSD-weighted integrand $S_{\delta x}(f) F(f, T)$, whose integral is the realized dephasing; its suppression ($\sim 5.8 \times 10^3$) is smaller than that of F alone ($\sim 2 \times 10^5$) because decoupling introduces a sensitivity feature near $f_{\text{drive}} = v/a$.

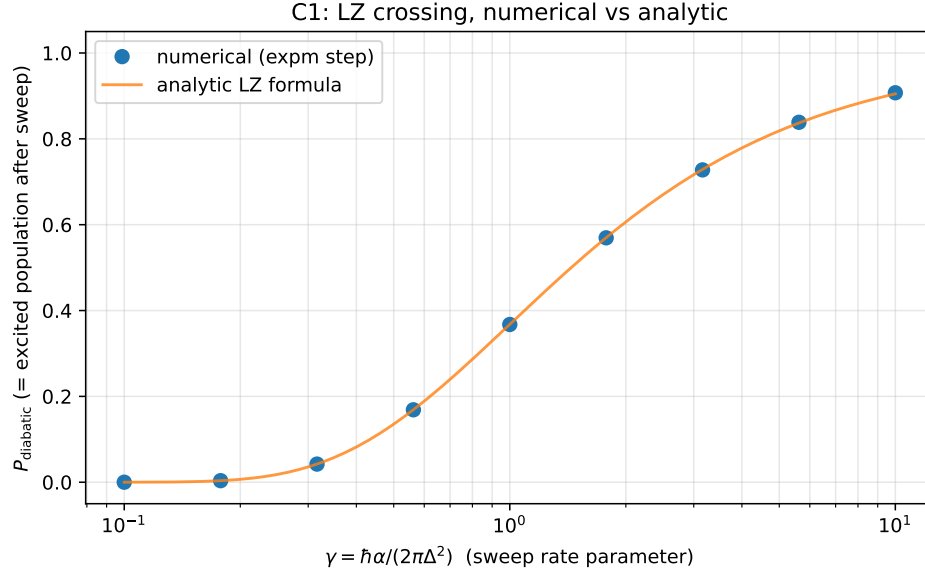


FIG. S3. Validation V4. Diabatic transition probability after a linear valley sweep, numerical (markers) versus the analytic Landau-Zener formula (line), as a function of the sweep-rate parameter $\gamma = \hbar\alpha/(2\pi\Delta_v^2)$. Agreement is within 0.28% across the scanned range.

the expected order-of-magnitude departure from the single-event formula, evidence that phase-coherent evolution between two crossings is captured.

E. Single-prism stray field (V5)

The analytic stray field of a uniformly magnetized rectangular prism (Engel-Herbert & Hesjedal) is implemented in JAX and matches the far-field dipole limit $B_z \rightarrow \mu_0 m / (2\pi d^3)$ to 0.06%: $8.95 \mu\text{T}$ (numerical) versus $8.96 \mu\text{T}$ (dipole) on the z -axis at $d = 5 \mu\text{m}$. Figure S4 shows the array field and its gradients at two depths.

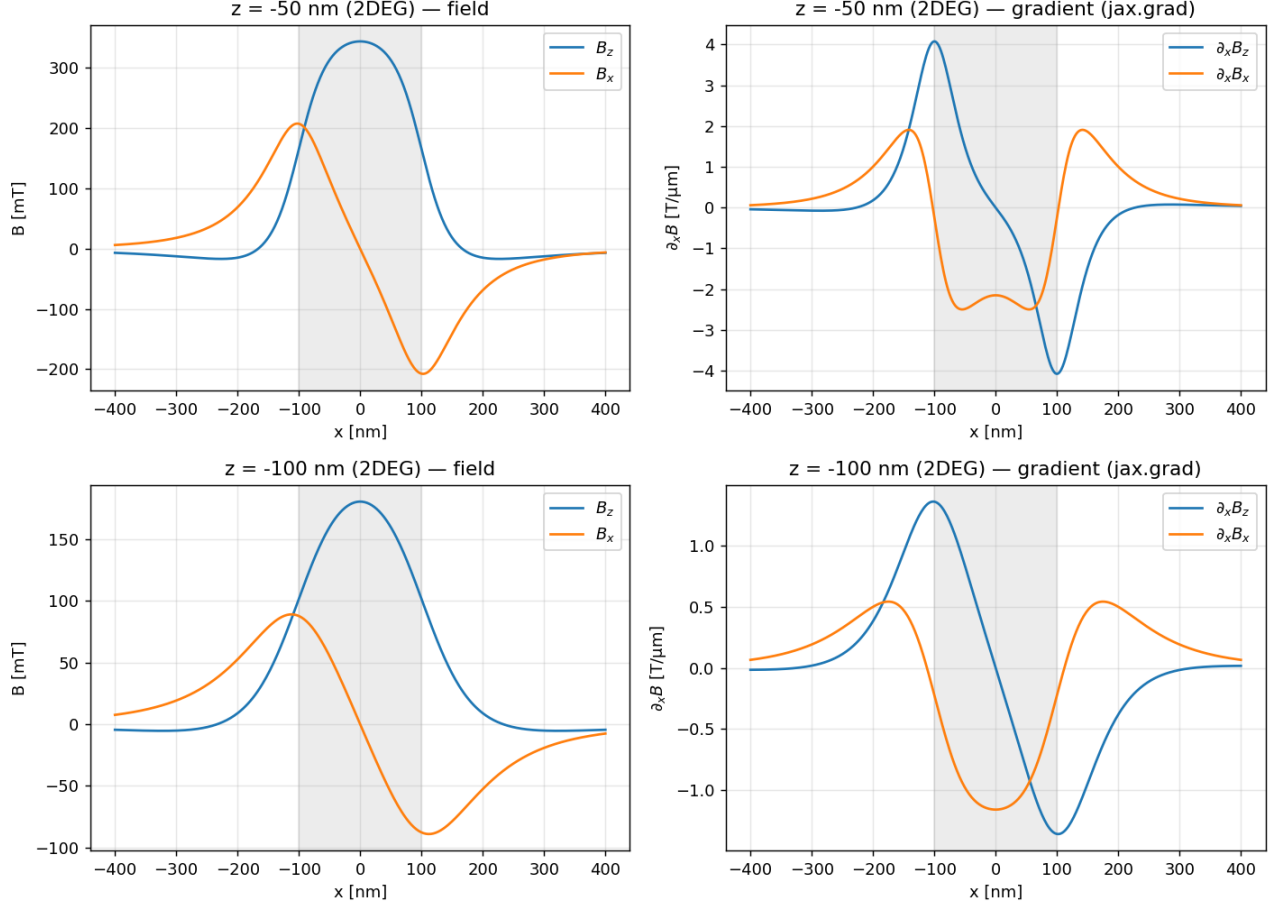


FIG. S4. Validation V5/V6. Stray field B_z, B_x (left) and their position gradients via automatic differentiation (right) at two depths below a micromagnet array. The gradient panels feed the coupling profiles of the main text; the autodifferentiation underlying them is validated against finite differences to $\leq 4 \times 10^{-6}$ (V6).

TABLE S2. Summary of literature and physical benchmark validations. Quantities validated as ratios or trends are marked; absolute magnitudes that depend on representative noise amplitudes are not claimed beyond the stated sense.

ID	quantity	reference / regime	result
V1	gradient-halving T_2^* ratio	Krzywda [1]: reported 1.93, our $\pm 20\%$ band	2.20 (ratio)
V2	motional-narrowing trend	Krzywda [1]: direction only	up to $64\times$
V3	filter-function suppression	static vs shuttling, this model	$F : 2 \times 10^5, \chi : 5.8 \times 10^3$
V4	valley Landau-Zener	reduced Oda model [2]	$< 0.28\%$
V5	single-prism stray field	dipole far-field limit	0.06%
V9	field/gradient scale	device parameter scales	comparable; peak gradient at upper end

F. Field- and gradient-scale consistency (V9)

For a $50 \times 50 \times 30$ nm magnet ($a = 150$ nm, depth -50 nm) the descriptor yields $\delta B_z = 26.9$ mT, peak gradient 1.19 mT/nm, and mean gradient 0.72 mT/nm. These are comparable to the micromagnet parameter scales $\delta B_z \sim 5$ – 30 mT and gradient ~ 0.1 – 1 mT/nm used in the main text and reported for comparable devices [3–5]: δB_z and the mean gradient fall inside those scales, and the peak gradient sits just above the upper end. This check bears on the field and gradient scales the chosen geometry produces, not on the magnet dimensions themselves; the velocity range is motivated separately by conveyor-mode shuttling studies. *Caveat:* a large magnet ($200 \times 200 \times 100$ nm) gives $\delta B_z \approx 197$ mT, about $10\times$ stronger; the main-text baseline uses the small-magnet configuration, and the large-magnet case is reserved as a strong-gradient stress test. Table S2 summarizes the checks of this section.

TABLE S3. Mean pairwise rank correlation under different definitions of the response magnitude. The ranking is high but weighting-dependent. The classification provides the decision-facing sign diagnostic, while the ranking provides complementary magnitude-ordering information.

magnitude definition	mean Spearman ρ
raw norm ($w_P=w_\chi=1$)	0.837
threshold-normalized	0.769
$ \Delta P_v $ only	0.751
$ \Delta \chi_\phi $ only	0.845
$P_{\text{ref}}=10^{-2}$, $\chi_{\text{ref}}=1$	0.754
$P_{\text{ref}}=10^{-4}$, $\chi_{\text{ref}}=10^{-3}$	0.769

TABLE S4. Mean classification agreement (exact agreement of the thresholded response label) under different effect-size floors ($|\Delta P_v|_{\text{min}}, |\Delta \chi_\phi|_{\text{min}}$), with one-channel cases separated.

floors	mean classification agreement
none	49.4%
$(10^{-5}, 10^{-4})$	36.3%
$(10^{-4}, 10^{-3})$ (main)	36.0%
$(10^{-3}, 10^{-2})$	58.0%

S3. ROBUSTNESS OF THE RESPONSE METRICS

The main-text conclusions rest on two derived quantities: the magnitude ranking (Spearman ρ) and the classification agreement. Here we check that the analysis choices do not drive the central finding: the thresholded classification is much more ansatz-dependent than magnitude ranking, with the lowest agreement in the A– B_x and B_z – B_x comparisons.

A. Ranking depends on channel weighting; the classification does not

The magnitude $|\mathcal{R}| = \sqrt{\Delta P_v^2 + \Delta \chi_\phi^2}$ weights the two channels equally in raw units, but $|\Delta \chi_\phi|$ is typically larger than $|\Delta P_v|$, so the raw-norm ranking is effectively dominated by the dephasing channel. Table S3 shows the mean rank correlation under several weightings: it ranges from $\rho \approx 0.75$ ($|\Delta P_v|$ -only) to $\rho \approx 0.85$ ($|\Delta \chi_\phi|$ -only), with the raw norm at 0.84. The ranking remains high but varies with channel weighting. We therefore use it as a complementary magnitude-ordering diagnostic. The classification depends only on which channels clear their floors and on their signs, so it is unaffected by this weighting choice.

B. Classification agreement is low across floors

The thresholded classification uses operational effect-size floors rather than statistical confidence bounds. Moving both floors a decade below and a decade above the main choice gives mean raw agreement between 36 and 58% (Table S4); the classification therefore remains far from model-independent throughout the sweep. The rise at the strictest floors should not be read as improved agreement among classified conditions: it is driven mainly by the two ansätze more often sharing the *below-threshold* label, which accounts for 336 of the 376 agreeing comparisons there. Restricted to comparisons in which both ansätze land in a two-channel quadrant, agreement is 44.8% (47/105) at the main floors and 31.8% (7/22) at the strictest floors. The qualitative conclusion, that the two-channel sign interpretation is poorly conserved across ansätze, is not tied to the particular floor choice. Candidate stability is assessed separately by the targeted $n_{\text{real}} = 30$ re-test in the main text. Figure S5 shows both sweeps.

C. Chance-corrected agreement is modest across pairs

Raw agreement can be inflated by skewed marginal distributions, and here the *below-threshold* category is the largest. Cohen’s κ is computed on the same nine-category label set and corrects for exactly that. It (Table S5) is modest throughout ($\kappa \approx 0.14$ – 0.33): no pair reaches strong agreement. The two lowest κ values are the B_z vs. B_x

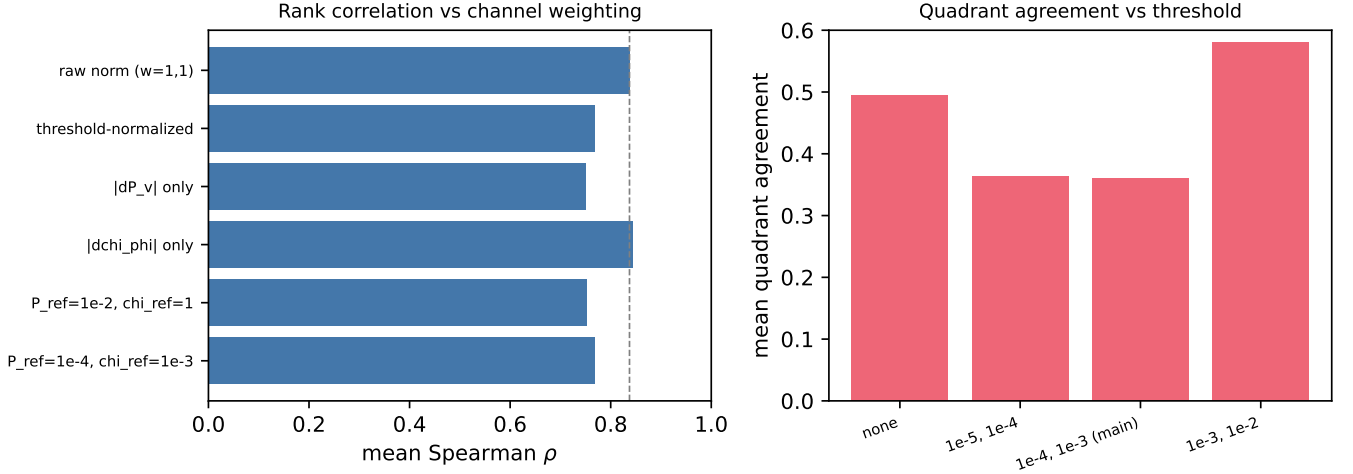


FIG. S5. Robustness of the two response metrics. Left: mean rank correlation under different channel weightings (dashed line: raw-norm value 0.837); the ranking is high but weighting-dependent. Right: mean classification agreement under different effect-size floors; it remains far from model-independent throughout the sweep. The archived right-panel title uses the earlier shorthand “quadrant agreement”; the plotted quantity is exact agreement of the full nine-category thresholded label.

TABLE S5. Pairwise classification agreement and chance-corrected Cohen’s κ at full validate (total-field convention), both over the nine-category thresholded label. Agreement is modest throughout; the two lowest values are the B_z/B_x and A/B_x pairs.

pair	raw agreement	Cohen’s κ
A vs A _{pocket}	36.1%	0.211
A vs B _z	45.4%	0.334
A vs B _x	29.6%	0.145
A _{pocket} vs B _z	31.5%	0.154
A _{pocket} vs B _x	44.4%	0.244
B _z vs B _x	28.7%	0.135

and A vs. B_x pairs, both pairing the $\partial_x B_x$ -based model with a $\partial_x B_z$ -based one. B_x is not a uniform outlier, however: the third such pair, A_{pocket} vs. B_x , has comparatively high κ ; this is consistent with the pocket-localized A_{pocket} sharing B_x ’s concentration near the flanks. The decisive axis is thus the gradient component modulated by spatial localization, consistent with the main-text finding, rather than a blanket property of the B_x ansatz.

D. Absolute performance proxies

The main text reports baseline-relative responses, not absolute error rates. For context, Table S6 lists the absolute full-model (M2) leakage P_v^{M2} , dephasing variance χ_ϕ^{M2} , and the derived phase-error proxy $p_\phi = (1 - e^{-\chi_\phi/2})/2$, averaged over the baseline geometry, both pocket cases, and $v \in \{5, 10, 20\}$ m/s. The absolute error scales are comparable across ansätze (same order of magnitude), confirming that the strong ansatz-dependence is in the *response interpretation*, not in a wildly different absolute performance.

S4. RELATION TO THE MAIN TEXT

The numerical-consistency checks (V0–V8) certify the field model, its gradients, and the time-evolution engine that produce every quantity in the main-text atlas. The literature and physical benchmark checks (V1–V9) support the representative parameter values (Table I of the main text). V9 verifies that the controlled-reference geometry produces field and gradient scales comparable to those of relevant micromagnet devices, while the velocity sweep is motivated separately by conveyor-mode shuttling studies. V1–V3 test the dephasing response to gradients, and V4 tests the reduced valley dynamics. None of these checks validates the spin–valley coupling profile $\lambda_{sv}(x)$ itself, which

TABLE S6. Absolute full-model performance proxies per ansatz, using the same setup as the atlas (baseline geometry, both pocket cases $x_c = 0$ and $x_c = 25$ nm, $v \in \{5, 10, 20\}$ m/s, $n_{\text{real}} = 5$, $1/f$ noise with $f_{\text{high}} = 10^7$ Hz). These are reported for context; the atlas does not optimize them.

ansatz	P_v^{M2}	χ_ϕ^{M2}	p_ϕ^{M2}
A	0.259	0.0195	0.0049
A _{pocket}	0.0786	0.0106	0.0026
B _z	0.266	0.172	0.0413
B _x	0.107	0.0169	0.0042

is treated in the main text as a phenomenological diagnostic ansatz rather than a microscopically derived coupling; the sensitivity of the conclusions to that choice is precisely the subject of the main text.

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